

Mathematics for Machine Learning

MAI391

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1. Analytic Geometry

1.1 Norm

Definition 1.1.1

A norm on V is a non-negative function

$$\|\cdot\| : V \rightarrow \mathbb{R} \quad (1.1)$$

such that $\forall k \in \mathbb{R}$ and $x, y \in V$:

- *Absolute homogeneous*: $\|kx\| = |k| \|x\|$
- **Triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$
- **Positive definite**: $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = \vec{0}$



Definition 1.1.2

Let p be a positive number. The l_p -norm $\|\cdot\|_p$ in $\mathbb{R}^n$:

$$\|x\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad \text{for any } x = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad (1.2)$$



Example 1.1.3

- $p = 1 \rightarrow$ Manhattan norm l_1 :

$$\|x\|_1 = \sum_{i=1}^n |x_i| \quad (1.3)$$

- $p = 2 \rightarrow$ Euclidean norm l_2 :

$$\|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2} \quad (1.4)$$

- $p = \infty \rightarrow$ maximum norm l_∞ :

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} \quad (1.5)$$



Definition 1.1.4

Euclidean norm on $\mathbb{R}^{m \times n}$:

$$\|A\| = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{\frac{1}{2}} \quad (1.6)$$

**Definition 1.1.5**

Spectral norm on $\mathbb{R}^{m \times n}$:

$$\|A\|_2 = \max_{x \in \mathbb{R}^n, \|x\|_2 = 1} \|Ax\|_2 \quad (1.7)$$



1.2 Inner Product

Definition 1.2.1

A linear mapping L on V is a function $L : V \rightarrow \mathbb{R}$ that satisfies:

- $L(kx) = kL(x)$
- $L(x + y) = L(x) + L(y)$

$\forall x \in V$ and $k \in \mathbb{R}$

**Definition 1.2.2**

A bilinear mapping or bilinear form B on V is a function $B : V \times V \rightarrow \mathbb{R}$ that is **linear in each argument separately**:

- $B(kx, z) = kB(x, z)$
- $B(x + y, z) = B(x, z) + B(y, z)$
- $B(x, kz) = kB(x, z)$
- $B(x, y + z) = B(x, y) + B(x, z)$

$\forall x, y, z \in V$ and $k \in \mathbb{R}$



Problem 1.2.3

Show that the mapping $B : R^2 \times R^2 \rightarrow R$

$$B(x, y) = x_1y_1 - 2x_1y_2 + 3x_2y_2 \quad (1.8)$$

is a bilinear mapping on R^2 .

◇

Solution.

$$\begin{aligned} B(kx, z) &= kx_1z_1 - 2kx_1z_2 + 3kx_2z_2 = k(x_1z_1 - 2x_1z_2 + 3x_2z_2) = kB(x, z) \\ B(x + y, z) &= (x_1 + y_1)z_1 - 2(x_1 + y_1)z_2 + 3(x_2 + y_2)z_2 \\ &= (x_1z_1 - 2x_1z_2 + 3x_2z_2) + (y_1z_1 - 2y_1z_2 + 3y_2z_2) \\ &= B(x, z) + B(y, z) \\ B(x, kz) &= kx_1z_1 - 2kx_1z_2 + 3kx_2z_2 = k(x_1z_1 - 2x_1z_2 + 3x_2z_2) = kB(x, z) \\ B(x, y + z) &= x_1(y_1 + z_1) - 2x_1(y_2 + z_2) + 3x_2(y_2 + z_2) \\ &= (x_1y_1 - 2x_1y_2 + 3x_2y_2) + (x_1z_1 - 2x_1z_2 + 3x_2z_2) \\ &= B(x, y) + B(x, z) \end{aligned}$$

$\Rightarrow B$ is linear mapping

□

Definition 1.2.4

A bilinear mapping $B : V \times V \rightarrow R$ is called

- **symmetric** if $B(x, y) = B(y, x)$
- **positive definite** if $B(x, x) > 0$ for any $V \ni x \neq \vec{0}$

♣

Definition 1.2.5

A positive definite, symmetric bilinear mapping $B : V \times V \rightarrow R$ is called an **inner product**. We typically write $\langle x, y \rangle$ instead of $B(x, y)$ for inner product

♣

Definition 1.2.6

The pair $(V, \langle \cdot, \cdot \rangle)$ is called **inner product space**

♣

Definition 1.2.7

The **dot product** or **scalar product** on $\mathbb{R}^n$ is given by

$$x^T y = \sum_{i=1}^n x_i y_i \quad (1.10)$$

**Property 1.2.8**

Dot product is inner product, and $(\mathbb{R}^n, \langle \cdot, \cdot \rangle)$ is **Euclidean vector space**

**Problem 1.2.9**

Consider $V = \mathbb{R}^2$. Define

$$\langle x, y \rangle = 2x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2 \quad (1.11)$$

Show that $\langle \cdot, \cdot \rangle$ is an inner product in $\mathbb{R}^2$



Solution.

- $\langle \cdot, \cdot \rangle$ is a bilinear mapping (*The solution is similar to Problem 1.2.3*)
- $\langle \cdot, \cdot \rangle$ is symmetric

$$\begin{aligned} \langle x, y \rangle &= 2x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2 \\ &= 2y_1 x_1 + y_1 x_2 + y_2 x_1 + 2x_2 y_2 = B(y, x) \end{aligned} \quad (1.12)$$

- $\langle \cdot, \cdot \rangle$ is positive definite

$$\begin{aligned} \langle x, x \rangle &= 2x_1^2 + 2x_2^2 + 2x_1 x_2 \\ &= (x_1 + x_2)^2 + x_1^2 + x_2^2 \geq 0 \end{aligned} \quad (1.13)$$

The equality occur only when $x_1 = 0, x_2 = 0, x_1 + x_2 = 0$ which is $x = \vec{0}$
 $\Rightarrow \langle \cdot, \cdot \rangle > 0$ for any $\mathbb{R}^2 \ni x \neq \vec{0}$, or $\langle \cdot, \cdot \rangle$ is positive definite

So $\langle \cdot, \cdot \rangle$ satisfies all properties to be an inner product

Definition 1.2.10

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is

- **positive definite** if

$$x^T A x > 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \quad (1.14)$$

- **positive semidefinite** if

$$x^T A x \geq 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \quad (1.15)$$

**Problem 1.2.11**

Consider the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \quad (1.16)$$

Show that A is positive definite but B is not



Solution.

$$\begin{aligned} x^T A x &= [x_1 \ x_2] \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= 2x_1^2 + 2x_1x_2 + 3x_2^2 \\ &> 0 \text{ for all non-zero vectors } x \in \mathbb{R}^n \end{aligned} \quad (1.17)$$

(the proof for why > 0 is similar to [Problem 1.2.9](#))

Also A is symmetric so A is positive definite

$$\begin{aligned} x^T B x &= [x_1 \ x_2] \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1^2 + 4x_1x_2 + 3x_2^2 \end{aligned} \quad (1.18)$$

Take $x = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow x^T B x = -1$

So B is not positive definite and not positive semidefinite

Note 1.2.12

When multiply matrices of form $x^T Ax$ (like in *Problem 1.2.11*), the coefficient of

$$x_i x_j = \begin{cases} a_{ij} , & \text{if } i = j \\ a_{ij} + a_{ji} , & \text{if } i \neq j \end{cases} \quad (1.19)$$

